

Applied Statistics

2025/2026, Summer

Course faculty	Ihor Miroshnychenko e-mail: imiroshnychenko@kse.org.ua
Office hours	on demand
ECTS credits	4
Duration	11 weeks
Class hours	20 lecture hours and 44 practice hours
Course venue	KSE Dragon Capital Building, 3 Mykola Shpak Street, Kyiv
Course language	Ukrainian/English

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Overview

Background and course rationale

In today's data-driven landscape, the ability to make confident, evidence-based decisions is a critical requirement across academia, business, and technology. While raw data is abundant, extracting actionable, reliable insights requires rigorous statistical analysis. Modern data scientists and analysts are frequently tasked with designing experiments, validating product changes (A/B testing), and interpreting complex metrics. However, there is often a significant gap between theoretical statistics taught in traditional settings and the practical, nuanced demands of real-world data, which is rarely perfectly normally distributed and often requires specialized techniques to detect subtle but impactful changes.

The Applied Statistics course is specifically designed to bridge the gap between statistical theory and real-world application. It provides a comprehensive, hands-on approach to experimental design and hypothesis testing, equipping students with the critical toolkit needed to tackle modern analytical challenges.

The curriculum is structured to progressively build analytical rigor:

- **Foundations of Inference (Weeks 1–4):** The course begins by solidifying core concepts of statistical inference, including significance, statistical power, minimum detectable effect (MDE), and classic parametric tests (Z-tests and T-tests).
- **Handling Real-World Data (Weeks 5–8):** Recognizing that real data is often messy or non-parametric, students will learn to robustly test for distributions and work with rank-based and frequency-based methods, including Goodness-of-Fit, Kolmogorov-Smirnov, Chi-square, and Mann-Whitney tests.
- **Advanced Industry Techniques (Weeks 9–10):** A distinguishing feature of this course is its deep dive into modern computation-heavy and variance-reduction techniques used by leading tech companies. Students will master Bootstrapping for complex confidence intervals, alongside cutting-edge methods for increasing test specificity, such as Ratio-metrics linearization, Stratification, and CUPED (Controlled Experiments Using Pre-Experiment Data).

By the end of the course, culminated in a final project presentation, students will not only understand how statistical tests work mathematically but will also know exactly when and why to apply them to maximize sensitivity and reliability in their analyses.

The course aims to

The course learning objectives are as follows:

1. **Build a Strong Foundation in Hypothesis Testing:** Enable students to deeply understand the mechanics of statistical tests, focusing on critical concepts such as statistical power, significance levels, confidence intervals, and Minimum Detectable Effect (MDE).
2. **Develop Practical Test Selection Skills:** Teach students how to critically evaluate data sets and select the most appropriate statistical methods—ranging from classic parametric tests (Z-tests, T-tests) to non-parametric alternatives (Mann-Whitney, Chi-square, Kolmogorov-Smirnov) based on specific data characteristics and hypotheses.
3. **Foster Rigor in Data Validation:** Provide the skills necessary to rigorously check statistical assumptions, including evaluating goodness-of-fit and testing for normality using tools like the Shapiro-Wilk test, Kolmogorov-Smirnov test, and QQ-plots.
4. **Introduce Industry-Standard Advanced Techniques:** Equip students with modern, computationally intensive methods used in advanced analytics and A/B testing, including Bootstrapping for complex distributions and variance reduction techniques like Stratification, Linearization, and CUPED.
5. **Enhance Analytical Communication:** Prepare students to independently design an analytical project, synthesize their statistical findings, and confidently present

actionable, data-driven conclusions to both technical and non-technical stakeholders.

Prerequisites

To succeed in this course, students should have:

- **Basic Probability and Descriptive Statistics:** Familiarity with fundamental concepts such as mean, median, variance, standard deviation, random variables, and basic probability distributions. While the course covers hypothesis testing from the ground up, a solid grasp of how to describe and summarize data is essential.
- **Programming Proficiency (Python or R):** As an applied course, students will be expected to implement statistical tests and advanced techniques (such as Bootstrapping and CUPED) programmatically. Basic proficiency in a language commonly used for data analysis - preferably Python (with libraries like pandas, numpy, and scipy) or R - is highly recommended.
- **Fundamental Data Literacy:** Prior experience with basic data manipulation, including loading datasets, filtering, grouping, and performing preliminary exploratory data analysis (EDA).
- **Basic Mathematical Foundations:** A general understanding of algebra and mathematical notation to comfortably follow and apply formulas related to statistical power, confidence intervals, and variance reduction techniques.

Learning outcomes

After completing this course, students will be able to:

1. Formulate and Evaluate Hypotheses: Define appropriate null and alternative hypotheses for various analytical scenarios, and correctly interpret p-values, significance levels, and confidence intervals.
2. Design Statistical Experiments: Calculate and apply statistical power and Minimum Detectable Effect (MDE) to determine appropriate sample sizes and design reliable experiments.
3. Select and Apply Statistical Tests: Identify and execute the correct parametric (Z-test, T-test) or non-parametric (Mann-Whitney, Chi-square) statistical test based on the specific characteristics of the data and the research question.
4. Validate Data Distributions and Assumptions: Assess data for normality and homogeneity using goodness-of-fit tests, the Kolmogorov-Smirnov test, the Shapiro-Wilk test, and visual diagnostics like QQ-plots.
5. Implement Resampling Methods: Apply Bootstrapping techniques to estimate distributions, construct robust confidence intervals, and perform two-sample comparisons when traditional parametric assumptions fail.

6. Optimize Test Sensitivity (A/B Testing): Apply advanced, industry-standard variance reduction techniques—such as Stratification, Linearization for ratio metrics, and CUPED—to increase the specificity and efficiency of statistical tests.
7. Communicate Analytical Findings: Design, conduct, and present an end-to-end applied statistics project, translating complex statistical results into clear, actionable insights for both technical and non-technical audiences.

Learning materials

All books are **freely available online**.

- [Introduction to Modern Statistics](#) (2e) by ine Çetinkaya-Rundel and Johanna Hardin
- [Statistical Inference via Data Science](#) by Chester Ismay and Albert Y. Kim
- [OpenIntro Statistics](#) by David M Diez, Christopher D Barr, Mine Çetinkaya-Rundel
- [Learning Statistics with R](#) by Danielle Navarro

Course Faculty



Ihor Miroshnychenko, PhD in Economics

 @araprof

 @datamirosh

 aranaur.rbind.io

 aranaur

 imiroshnychenko@kse.org.ua

 ihormiroshnychenko

Career Highlights:

- Teacher, Faculty of Computer Science, KSE
- [Associate Professor](#), Department of Technology Management, FIT, KNU
- [Lecturer](#), MBA, International Institute of Business

Research Interests:

- Data Science
- Statistics
- Data Analysis
- Machine Learning

Teaching Philosophy:

I believe that statistics is a powerful tool for understanding the world around us. My goal is to help students develop a deep appreciation for the role of data in decision-making and to equip them with the skills to analyze data effectively and communicate their findings clearly.

Personal Interests:

- Guitar playing
- Traveling
- Photography
- Gaming
- Reading
- Football

Interesting Facts:

- Big fan of the Middle-Earth universe.
- Engaged in historical reconstruction
- Metalhead

Course Structure

The course consists of 10 lectures and 10 labs.

The goal of the lectures and labs is to make them as interactive as possible. My role as instructor is to introduce you to new tools and techniques, but it is up to you to take them and use them. A lot of what you do in this course will involve writing code, and coding is a skill that is best learned by doing. Therefore, as much as possible, you will work on various tasks and activities throughout each lab. Attendance will not be taken during class, but you are expected to attend all lectures and lab sessions and meaningfully contribute to in-class exercises and discussions.

You are expected to bring a laptop to each class to participate in in-class exercises using the R or Python programming languages.

The next table outlines the semester's topics and content. Note that this schedule will be updated as the semester progresses, and the timelines for topics and assignments may change throughout the semester.

WEEK	TOPIC	SUBTOPIC
1	Introduction to Statistical Tests	What is a statistical test? Significant level Types of statistical tests
2	Statistical Power and Sensitivity	What is power? Minimum detectable effect Confidence intervals
3	The Z-Test	Comparing the mean Central limit theorem Continuity correction
4	The T-Test	Why t and not the z-test? t-test and CI MDE for t-test
5	Goodness-of-Fit and Normality	Goodness-of-fit test. Kolmogorov's test Testing for normality. Shapiro-Wilk test. QQ-plot
6	Kolmogorov-Smirnov Test	Kolmogorov-Smirnov homogeneity test Sample size of the Kolmogorov-Smirnov criterion Independence of sampling elements
7	Chi-Square Test	Pearson's Chi-square test Pearson's Chi-square test for independence Pearson's Chi-square test for homogeneity Yates correction
8	Mann-Whitney test	Mann-Whitney test Alternative hypothesis of the Mann-Whitney criterion One-sided Mann-Whitney criterion Does the Mann-Whitney work for other hypotheses?
9	Bootstrapping	

WEEK	TOPIC	SUBTOPIC
		Introduction to Bootstrap
		Two-sample bootstrap
		Types of bootstrap confidence intervals
		Bootstrap vs. T-test
10	Linearization and Increasing Test Specificity	
		Ratio-metrics and bootstrap
		Ratio-metrics and linearization
		Stratification
		CUPED
11	Project Presentation	

Grading Policy

The final course grade will be calculated as follows:

Category	Percentage	Description
Practical Labs	80%	10 lab assignments (8 points each)
Final Project & Presentation	20%	Final team project and presentation (20 points)
Class Participation & Attendance	5%	Engagement in class discussions (bonus-based)

Class Participation & Attendance (5% Bonus)

Active engagement in lectures, discussions, and peer reviews during the final presentations. This includes:

- Regular attendance
- Voluntary contributions to discussions
- Constructive peer interaction during lab work
- Asking meaningful questions or offering insights related to course content

Bonus points will be added to your final grade (up to 5 points). Consistent inactivity or disengagement may result in 0 bonus points.

Labs (80%)

There will be **10 lab assignments**, each worth **up to 8 points**, aligned with the key topics of the course. Labs are designed to:

- Apply theoretical concepts in practice

- Develop proficiency with statistical software (R or Python)
- Practice data wrangling, visualization, and interpretation of results

Lab grading criteria per assignment:

Criterion	Points
Correctness of methodology	3
Interpretation of results	3
Code quality and organization	2
Total	8

Labs must be submitted in the specified format (PDF, HTML). While collaboration is allowed, all submitted work must reflect **your own understanding** and be **written in your own words**.

Project (20%)

The course project is a team-based data analysis project. It involves choosing a dataset, formulating a research question, applying appropriate statistical methods, and presenting your findings.

You must complete the project to pass the course.

The final grade (20 points) will be distributed across the following components:

Criterion	Points
Data preparation	4
Visualization and graphics	4
Interpretation of results	4
Report quality (clarity, structure, references)	4
Code quality and reproducibility	4
Total	20

The final deliverables include:

- A written report
- Reproducible code (submitted via GitHub or archive)

- A live project presentation (in class)

More detailed instructions and rubrics will be provided mid-semester.

Five tips for success

Your success in this course depends crucially on you and the effort you put into it. The course has been organized so that the burden of learning is on you. I will help you by providing you with materials, answering questions, and setting a pace, but for this to work, you must do the following:

1. Ask questions. As often as you can. In class, out of class. Ask me, ask your friends, ask the person sitting next to you. This will help you more than anything else. If you're not sure about the lab, ask. If the reading is confusing, ask
2. Do the readings.
3. Do the labs. The earlier you start, the better. It's not enough to just mechanically plow through the exercises. You should ask yourself how these exercises relate to earlier material, and imagine how they might be changed.
4. Don't procrastinate. The content builds on what was taught in previous weeks, so if something is confusing to you in Week 2, Week 3 will be even more confusing, Week 4 even worse, etc.

Academic integrity warning

Academic integrity is highly valued by the KSE community and its vast majority of students. We have a zero-tolerance policy towards academic plagiarism, self-plagiarism, fabrication, falsification, cheating, deception, bribery and other types of academic integrity violations. Due to the highly competitive nature of the programmes, all students must be treated equally. Even a single case of violation of academic integrity is a serious misdemeanour that may lead to unjust redistribution of grades and, consequently, overall rankings, possible grants, fee reductions, and other merit-based awards. Therefore, penalties may vary from receiving zero points or a negative grade for an assignment to expulsion of the student from KSE, depending on the severity of the case and might include additional consequences such as deprivation of scholarships, financial assistance etc. All the rules and procedures regarding academic integrity are stated in the KSE Code of Academic Integrity. Before enrolling in this course, students must be aware of and abide by the KSE Code of Academic Integrity.

Academic honesty

TL;DR: Don't cheat!

Please abide by the following as you work on assignments in this course:

- **Collaboration:** Only work that is clearly assigned as team work should be completed collaboratively.
 - You may discuss lab assignments with other students; however, you may not directly share (or copy) code or write up with other students. For team assignments, you may collaborate freely within your team. You may discuss the assignment with other teams; however, you may not directly share (or copy) code or write up with another team. Unauthorized sharing (or copying) of the code or write up will be considered a violation for all students involved.
 - For the project, collaboration within teams is not only allowed, but expected. Communication between teams at a high level is also allowed however you may not share code or components of the project across teams.
- **Online resources:** I am well aware that a huge volume of code is available on the web to solve any number of problems. Unless I explicitly tell you not to use something, the course's policy is that you may make use of any online resources (e.g., StackOverflow) but you must explicitly cite where you obtained any code you directly use (or use as inspiration). Any recycled code that is discovered and is not explicitly cited will be treated as plagiarism.
- **Use of generative artificial intelligence (AI):** You should treat generative AI, such as ChatGPT, the same as other online resources. There are two guiding principles that govern how you can use AI in this course:
 1. Cognitive dimension: Working with AI should not reduce your ability to think clearly. We will practice using AI to facilitate—rather than hinder—learning.
 2. Ethical dimension: Students using AI should be transparent about their use and make sure it aligns with academic integrity.
 -  AI tools for code: You may make use of the technology for coding examples on assignments; if you do so, you must explicitly cite where you obtained the code. Any recycled code that is discovered and is not explicitly cited will be treated as plagiarism. You may use these guidelines for citing AI-generated content.
 -  AI tools for narrative: Unless instructed otherwise, you may not use generative AI to write narrative on assignments. In general, you may use generative AI as a resource as you complete assignments but not to answer the exercises for you.

You are ultimately responsible for the work you turn in; it should reflect your understanding of the course content.

Late work & extensions

The assignment due dates are intended to help you keep up with the course material and to ensure the teaching team can provide feedback promptly. We understand that

things come up from time to time that can make it challenging to submit an assignment by the deadline.

- Labs may be submitted up to 3 days late. Two points will be deducted for every 24 hours the assignment is late.
- The late work policy for the project will be provided with the project instructions.